Model vs Index — CAGR Comparison

2020-07-01 to 2025-01-31 | monthly rebalance, 52 periods | equal-weighted top 20 | 1-month holding, non-overlapping | generated 2026-08-20T11:56:28

Basis

Price return compared with price return. NSE's historical index endpoint serves price indices only — every total-return (TRI) spelling returns no data. The strategy's dividends are therefore reported separately rather than folded into the headline, because comparing a dividend-inclusive strategy against a dividend-less index would credit the model with roughly the market's yield (about 1–1.5% a year in India) for nothing.

Index returns are measured over the sessions the strategy actually traded — the same next-session entry and the same exit — not between month-ends, so neither side is measuring a different span of market time.

Headline: CAGR by strategy

Strategy	Gross CAGR	Net CAGR	vs N50	vs N500	vs NMIDCAP 150	vs NSMALLCAP 250
equal weight universe	42.41%	40.92%	**+23.11%**	**+20.14%**	**+10.55%**	**+8.35%**
growth only	49.74%	46.27%	**+28.45%**	**+25.49%**	**+15.90%**	**+13.69%**
model 5	69.38%	60.12%	**+42.31%**	**+39.34%**	**+29.76%**	**+27.55%**
momentum only	57.62%	43.01%	**+25.20%**	**+22.23%**	**+12.64%**	**+10.44%**
quality only	33.99%	32.15%	**+14.34%**	**+11.37%**	**+1.79%**	**-0.42%**
random ranking	38.07%	11.76%	**-6.05%**	**-9.02%**	**-18.61%**	**-20.81%**
simple quality momentum	38.65%	30.10%	**+12.28%**	**+9.31%**	**-0.27%**	**-2.48%**
value only	93.99%	86.61%	**+68.80%**	**+65.83%**	**+56.25%**	**+54.04%**

vs <index> is the strategy's CAGR minus that index's CAGR over the identical periods, net of costs where a cost model ran. A positive number is outperformance in annualised percentage points.

Index reference

Index	CAGR	Volatility	Sharpe	Max drawdown	Periods
NIFTY 50	17.81%	19.56%	0.93	-12.74%	52
NIFTY 500	20.78%	20.74%	1.01	-16.81%	52
NIFTY MIDCAP 150	30.37%	25.20%	1.17	-19.71%	52
NIFTY SMALLCAP 250	32.57%	29.32%	1.10	-25.36%	52
Eligible universe (equal weight)	42.41%	-	-	-	52

The **eligible universe** row is the benchmark that controls for size and breadth: the same point-in-time universe the strategy chose from, held equally weighted. Beating NIFTY 500 while trailing this means the model captured a small-cap tilt rather than stock selection.

Risk-adjusted comparison

Strategy	Basis	CAGR	Volatility	Sharpe	Max DD	Hit rate
equal weight universe	net	40.92%	30.62%	1.27	-24.66%	67%
growth only	net	46.27%	30.78%	1.40	-23.03%	67%
model 5	net	60.12%	30.69%	1.70	-23.07%	69%
momentum only	net	43.01%	36.92%	1.16	-29.87%	63%
quality only	net	32.15%	23.45%	1.31	-22.31%	67%
random ranking	net	11.76%	36.52%	0.47	-40.78%	62%
simple quality momentum	net	30.10%	30.99%	1.01	-28.60%	63%
value only	net	86.61%	42.75%	1.68	-34.02%	69%

Excess return against NIFTY 500

Strategy	Mean excess/period	Tracking error	Information ratio	Periods beaten
equal weight universe	+1.50%	16.77%	1.07	65%
growth only	+1.84%	20.30%	1.09	60%
model 5	+2.61%	20.96%	1.49	63%
momentum only	+1.81%	27.37%	0.79	60%
quality only	+0.82%	13.75%	0.71	56%
random ranking	-0.32%	22.86%	-0.17	48%
simple quality momentum	+0.85%	22.73%	0.45	52%
value only	+4.23%	29.62%	1.71	67%

Information ratio is excess return per unit of tracking error — how reliably the outperformance was earned, not how large it was. **Periods beaten** is the share of rebalances where the strategy outpaced the index; a high CAGR with a low share was carried by a handful of periods.

Turnover and cost drag

Strategy	One-way turnover	Gross CAGR	Net CAGR	Cost drag
equal weight universe	0.058	42.41%	40.92%	1.49%
growth only	0.122	49.74%	46.27%	3.47%
model 5	0.362	69.38%	60.12%	9.25%
momentum only	0.486	57.62%	43.01%	14.62%
quality only	0.086	33.99%	32.15%	1.83%
random ranking	0.978	38.07%	11.76%	26.31%
simple quality momentum	0.419	38.65%	30.10%	8.55%
value only	0.173	93.99%	86.61%	7.38%

Cost drag is annualised, and it is the difference between a ranking that works on paper and one that works in an account. A strategy replacing most of its book every month pays it repeatedly.

Before quoting these numbers

An index is investable and this portfolio is a simulation. The index CAGR is achievable through a low-cost fund; the strategy CAGR assumes every fill happened at the modelled price and size.

The window is what it is. A CAGR over a few years is dominated by the market regime inside it. Check the periods-beaten share and the drawdown before treating a CAGR gap as skill.

Costs are modelled, not measured. Statutory charges are exact and effective-dated; half-spread and market impact are assumptions, which is why gross and net are both shown.

Delisting recovery is an assumption. Re-run with --delisting-strategy zero and last_close to see how much of the gap depends on it.